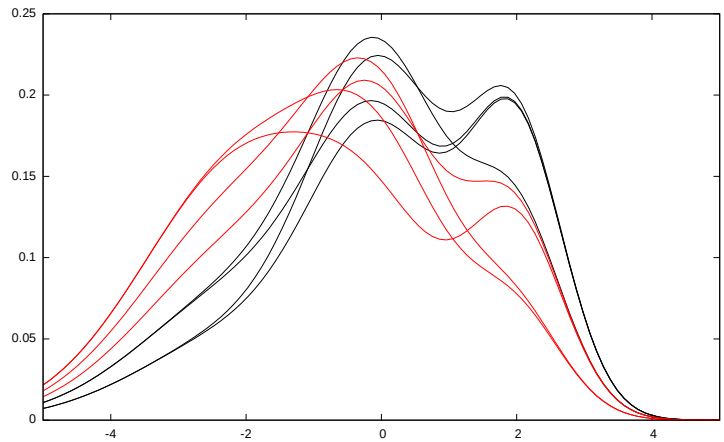

Figure 15: Example of w-GMM clustering into $k = 2$ clusters.

on a mixture family (structure $(\mathcal{M}, \mathrm{KL})$). There are many other dually flat structures that can be met in a statistical context: For example, two other dually flat structures for the D-dimensional probability simplex Δ_D are reported in Amari's textbook [8]: (1) the conformally deforming of the α-geometry (page 88, Eq. 4.95 of [8]), and (2) the χ-escort geometry (page 91, Eq. 4.114 of [8]).

5 Conclusion: Summary, historical background, and perspectives

5.1 Summary

We explained the dualistic nature of information manifolds (M, g, ∇, ∇^*) in information geometry. The dualistic structure is defined by a pair of conjugate connections coupled with the metric tensor that provides a dual parallel transport that preserves the metric. We showed how to extend this structure to a 1-parameter family of structures: From a pair of conjugate connections, the pipeline to build this 1-parameter family of structures can be informally summarized as:

$$(M, g, \nabla, \nabla^*) \Rightarrow (M, g, C) \Rightarrow (M, g, \alpha C) \Rightarrow (M, g, \nabla^{-\alpha}, \nabla^{\alpha}), \quad \forall \alpha \in \mathbb{R}. \tag{202}$$

We stated the fundamental theorem of information geometry on dual constant-curvature manifolds, including the special but important case of dually flat manifolds on which there exists two potential functions and global affine coordinate systems related by the Legendre-Fenchel transformation. Although, information geometry historically started with the Riemannian modeling $(\mathcal{P}, {}_{\mathcal{P}}g)$ of a parametric family of probability distributions $\mathcal{P}$ by letting the metric tensor be the Fisher information matrix, we have emphasized the dualistic view of information geometry which considers non-Riemannian manifolds that can be derived from any divergence, and not necessarily tied to a statistical context (e.g., information manifold can be used in mathematical programming [101]). Let us notice that for any symmetric divergence (e.g. any symmetrized f-divergence like the squared Hellinger divergence), the induced conjugate connections coincide with the Levi-Civita connection but the Fisher-Rao metric distance does not coincide with the squared Hellinger divergence.

On one hand, a Riemannian metric distance D_ρ is *never* a divergence because the rooted distance functions fail to be smooth at the extremities but a squared Riemmanian metric distance is always a divergence. On the other hand, taking the power δ of a divergence D (i.e., D^δ) for some $\delta > 0$ may yield a metric distance (e.g., the square root of the Jensen-Shannon divergence [44]), but this may not always be the case: The powered Jeffreys divergence J^δ is never a metric distance (see [127], page 889). Recently, the *Optimal Transport*

(OT) theory [130] gained interest in statistics and machine learning. But the optimal transport between two members of a same elliptically-contoured family has the same optimal transport formula distance (see [40] Eq. 16 and Eq. 17, although they have different Kullback-Leibler divergences). Another essential difference is that the Fisher-Rao manifold of location-scale families is hyperbolic but the Wasserstein manifold of location-scale families has positive curvature [40, 125].

Notice that we may convert back and forth a similarity $S(p,q) \in (0,1]$ to a dissimilarity $D(p,q) \in [0,\infty)$ as follows:

$$S(p,q) = \exp(-D(p,q)) \in (0,1] \quad (203)$$

$$D(p,q) = -\log S(p,q) \in [0,\infty) \quad (204)$$

When the dissimilarity satisfies the (additive) triangle inequality (i.e., $D(p,q)+D(q,r) \geq D(p,r)$ for any triple (p,q,r)) then the corresponding similarity satisfies the multiplicative triangle inequality: $S(p,q) \times S(q,r) \leq S(p,r)$. A metric transform on a metric distance D is a transformation T such that $T(D(p,q))$ is a metric. The transformation $T(u) = \frac{1}{1+u}$ is a metric transform which bounds potentially unbounded metric distances: That is, if D is an unbounded metric, then $T(D(p,q)) = \frac{D(p,q)}{1+D(p,q)}$ is a bounded metric distance. The transformation $S(u) = u^2$ is not a metric transform since the squared of the Euclidean metric distance is not a metric distance.

5.2 A brief historical review of information geometry

The field of Information Geometry (IG) was historically motivated by providing some differential-geometric structures to statistical models in order to reason geometrically about statistical problems with the endeavor goal of geometrizing mathematical statistics [29, 5, 39, 67, 54, 58, 9, 47]: Professor Harold Hotelling [53] first considered in the late 1920's the *Fisher Information Matrix* (FIM) I as a Riemannian metric tensor g (ie., the *Fisher Information metric*, FIm), and interpreted a parametric family of probability distributions M as a Riemannian manifold (M,g). Historically speaking, Hotelling attended the American Mathematical Society's Annual Meeting in Bethlehem (Pennsylvania, USA) on December 26–29, 1929, but left before his scheduled talk on December 27. His handwritten notes on the "Spaces of Statistical Parameters" was read by a colleague and are fully typeset in [123]. We warmly thank Professor Stigler for sending us the scanned handwritten notes and for discussing by emails some historical aspects of the birth of information geometry. In this pioneering work, Hotelling mentioned that location-scale probability families yield Riemannian manifolds of constant non-positive curvatures. This Riemannian modeling of parametric family of densities was further independently studied by Calyampudi Radhakrishna Rao (C.R. Rao) in his celebrated paper [111] (1945) that also includes the Cramér-Rao lower bound [71] and the Rao-Blackwellization technique used in statistics. Nowadays the induced Riemannian metric distance is often called the *Fisher-Rao distance* [122] or Rao distance [113]. Yet another use of Riemannian geometry in statistics was pioneered by Harold Jeffreys [55] that proposed to use as an invariant prior the normalized volume element of the Fisher-Riemannian manifold. In those seminal papers, there was no theoretical justification of using the Fisher information matrix as a metric tensor (besides the fact that it is a well-defined positive-definite matrix for regular identifiable models). Nowadays, this Riemmanian metric tensor is called the *information metric* for short. Modern information geometry considers a generalization of this approach using a non-Riemannian dualistic modeling (M,g,∇,∇^*) which coincides with the Riemannian manifold when $\nabla = \nabla^* = {}^{\mathrm{LC}}\nabla$, the Levi-Civita connection (the unique torsion-free affine connection compatible with the metric tensor). The Fisher-Rao geometry has also been explored in thermodynamics yielding the Ruppeiner geometry [134], and the geometry of thermodynamics is called nowadays called *geometrothermodynamics* [109].

In the 1960's, Nikolai Chentsov (also commonly written Čencov) studied the algebraic category of all statistical decision rules with its induced geometric structures: Namely, the α-geometries ("equivalent differential geometry") and the dually flat manifolds ("Nonsymmetric Pythagorean geometry" of the exponential families with respect to the Kullback-Leibler divergence). In the preface of the english translation of his 1972's russian monograph [29], the field of investigation is defined as "geometrical statistics." However in the original Russian monograph, Chentsov used the russian term *geometrostatistics*. According to Professor

Alexander Holevo, the geometrostatistics term was coined by Andrey Kolmogorov to define the field of differential geometry of statistical models. In the monograph of Chentsov [29], the Fisher information metric is shown to be the unique metric tensor (up to a scaling factor) yielding statistical invariance under Markov morphisms (see [26] for a simpler proof that generalizes to positive measures).

The dual nature of the information geometry was thoroughly investigated by Professor Shun-ichi Amari [4]. In the preface of his 1985's monograph [5], Professor Amari coined the term *information geometry* as follows: "The differential-geometrical method developed in statistics is also applicable to other fields of sciences such as information theory and systems theory... They together will open a new field, which I would like to call *information geometry*." Professor Amari mentioned in [5] that he considered the Gaussian Riemannian manifold as a hyperbolic manifold in 1959, and was strongly influenced by Efron's paper on statistical curvature [41] (1975) to study the family of α-connections in the 1980's [4, 68]. Professor Amari prepared his PhD under the supervision of Professor Kondo [31], an expert of differential geometry in touch with Professor Kawaguchi [59]. The role of differential geometry in statistics has been discussed in [17].

Note that the dual affine connections of information geometry have also been investigated independently in *affine differential geometry* [99] which considers invariance under volume-preserving affine transformations by defining a volume form instead of a metric form for Riemannian geometry. The notion of dual parallel transport compatible with the metric is due to Aleksandr Norden [100] and Rabindra Nath Sen [116, 117, 118] (See the Senian geometry in `http://insaindia.res.in/detail/N54-0728`).

We summarize the main fundamental structures of information manifolds below:

(M, g)	Riemannian manifold
$(\mathcal{P}, {}_{\mathcal{P}}g)$	Fisher-Riemannian (expected) Riemannian manifold
(M, g, ∇)	Riemannian manifold (M, g) with affine connection ∇
$(\mathcal{P}, {}_{\mathcal{P}}g, {}_{\mathcal{P}}{}^{e}\nabla^{\alpha})$	Chentsov's manifold with affine exponential α-connection
(M, g, ∇, ∇^*)	Amari's dualistic information manifold
$(\mathcal{P}, {}_{\mathcal{P}}g, {}_{\mathcal{P}}\nabla^{-\alpha}, {}_{\mathcal{P}}\nabla^{\alpha})$	Amari's (expected) information α-manifold, α-geometry
(M, g, C)	Lauritzen's statistical manifold [62]
$(M, {}^{D}g, {}^{D}\nabla, {}^{D^*}\nabla)$	Eguchi's conjugate connection manifold induced by divergence D
$(M, {}^{F}g, {}^{F}C)$	Chentsov/Amari's dually flat manifold induced by convex potential F

We use the $\equiv$ symbol to denote the equivalence of geometric structures. For example, we have $(M, g) \equiv (M, g, {}^{\mathrm{LC}}\nabla, {}^{\mathrm{LC}}\nabla^* = {}^{\mathrm{LC}}\nabla)$.

5.3 Perspectives

We recommend the two recent textbooks [25, 8] for an indepth covering of (parametric) information geometry, and the book [48] for a thorough description of some infinite-dimensional statistical models. (Japanese readers may refer to [7, 45].) We did not report the various coefficients of the metric tensors, Christoffel symbols and skewness tensors for the expected α-geometry of common parametric models like the multivariate Gaussian distributions, the Gamma/Beta distributions, etc. They can be found in [10, 25] and in various articles dealing with less common family of distributions [64, 65, 141, 105, 142, 104, 10]. Although we have focused on the finite parametric setting, information geometry is also considering non-parametric families of distributions [107], and quantum information geometry [51].

We have shown that we can always create an information manifold (M, D) from *any* divergence function D. It is therefore important to consider generic classes of divergences in applications, that are ideally axiomatized and shown to have exhaustive characteristics. Beyond the three main Bregman/Csiszár/Jensen classes (theses classes overlap [102]), we may also mention the class of conformal divergences [98, 91], the class of projective divergences [92, 97], etc. Figure 16 illustrates the relationships between the principal classes of distances.

There are many perspectives on information geometry as attested by the new Springer journal[13], and the biannual international conference "Geometric Sciences of Information" (GSI) [80, 81, 82] with its collective

[13] 'Information Geometry', `https://www.springer.com/mathematics/geometry/journal/41884`